

KMRL Signalling Systems Audit · Q2 FY2025-26

Kochi Metro Rail Limited

Directorate of Signalling & Telecommunications

Quarterly Audit Report ? Q2 FY2025-26

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Classification: Internal ? Engineering

1. SCOPE

This audit covers the Communications-Based Train Control (CBTC) installation on the Aluva-Pettah corridor, comprising 22 stations and 25.6 route kilometres. The review period runs from 01 July 2025 to 30 September 2025.

2. SUMMARY OF FINDINGS

During the review period, 12 intermittent Automatic Train Protection (ATP) faults were recorded. All 12 events were traced to a firmware mismatch on rev 4.2 On-Board Units (OBUs) fitted to trainsets TS-04, TS-08 and TS-11. The fault manifests as a spurious emergency brake application when the OBU hands over between radio block centres near Edappally station.

No passenger injuries resulted from any of the 12 events. Average service delay per event was 4.5 minutes.

3. ROOT CAUSE

The rev 4.2 OBU firmware implements an older handover timer (1200 ms) than the wayside equipment expects (800 ms). Under degraded radio conditions the timer expires before handover completes and the OBU falls back to a safe-brake state.

Remediation and Contractual Position

4. REMEDIATION

Patch v4.3 corrects the handover timer to 800 ms and has completed factory acceptance testing at the vendor's Bengaluru facility. Deployment to the fleet is currently BLOCKED pending resolution of the Alstom Annual Maintenance Contract, which expires on 31 December 2025.

Clause 11.3 of the AMC governs renewal and continuity obligations. Until the contract is renewed, the vendor will not certify firmware releases for revenue service.

5. RECOMMENDATIONS

R1. Accelerate the AMC continuity decision to the November 2025 board meeting.

Owner: Directorate of Procurement. Priority: HIGH.

R2. Deploy firmware v4.3 to all rev 4.2 OBUs in the first maintenance window following AMC renewal. Owner: Directorate of S&T. Priority: HIGH.

R3. Until v4.3 is deployed, impose a temporary 40 kmph speed restriction through the Edappally radio block boundary. Owner: Operations Control Centre. Priority: MEDIUM.

R4. Extend OBU event logging retention from 30 days to 180 days. Priority: LOW.

6. FINANCIAL IMPACT

Estimated cost of the AMC renewal is INR 4.2 crore per annum. Estimated revenue loss attributable to the 12 ATP events is INR 6.8 lakh.

7. NEXT REVIEW

The next audit is scheduled for Q3 FY2025-26, week commencing 05 January 2026.